Lab 12: The Muscular System

BIOL-8

Name: _____ **Date:** _____

Lab Section: _____

Learning Objectives

By the end of this lab, you will be able to:

1. **Compare and contrast** the three types of muscle tissue by location, structure, and control.
2. **Describe the sliding filament mechanism** of skeletal muscle contraction, tracing the role of calcium, troponin, tropomyosin, actin, and myosin.
3. **Explain the neuromuscular junction** — how a motor neuron triggers an action potential and calcium release in a muscle fiber.
4. **Distinguish motor units and fiber types** (slow-twitch vs. fast-twitch) and predict how each contributes to force and fatigue.
5. **Differentiate isotonic and isometric contractions** and identify real-world examples of each.
6. **Interpret experimental data** from a grip fatigue test to draw conclusions about muscle energy systems and fatigue mechanisms.

Overview

Muscles are responsible for every movement your body makes — from blinking to running a marathon. They also generate most of your body heat and stabilize every joint you have. Today you will review the major types of muscle tissue, explore the physiology of skeletal muscle contraction, and then perform a **grip fatigue test** to investigate muscle endurance and fiber type recruitment.

Materials: Stress ball (or tennis ball) · Timer · This worksheet

Duration: ~1 hour

Background: Three Types of Muscle Tissue

Feature	Skeletal Muscle	Cardiac Muscle	Smooth Muscle
Location	Attached to bones	Heart wall	Walls of hollow organs (stomach, blood vessels, bladder)
Appearance	Striated (striped)	Striated	Non-striated
Control	Voluntary	Involuntary	Involuntary
Cell shape	Long, cylindrical, multinucleate	Branched, intercalated discs	Spindle-shaped, uninucleate
Speed	Fast to slow (fiber type)	Rhythmic, sustained	Slow, sustained
Example action	Picking up a cup	Pumping blood	Moving food through intestines

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Background: How Skeletal Muscle Contracts

Structural Organization

Skeletal muscles are organized from large to small:

> ***Muscle → Fascicle → Muscle fiber (cell) → Myofibril → Sarcomere***

Each **muscle fiber** is a single, long, multinucleate cell packed with **myofibrils** — bundles of contractile proteins organized into repeating units called **sarcomeres**.

The Sarcomere and Sliding Filament Theory

The **sarcomere** is the functional unit of contraction. It contains two interleaved sets of protein filaments:

- **Thick filaments** — myosin, with projecting "heads" that form **cross-bridges** with actin
- **Thin filaments** — actin, coated by the regulatory proteins **troponin** and **tropomyosin**

How a contraction happens (step by step)

1. A nerve signal arrives → ACh is released at the neuromuscular junction
2. ACh binds receptors on the muscle fiber → action potential travels along the sarcolemma and into T-tubules
3. T-tubules signal the **sarcoplasmic reticulum** to release Ca^{2+} into the cytoplasm

4. Ca^{2+} binds **troponin**, which shifts **tropomyosin** away from myosin-binding sites on actin
5. Myosin heads attach to actin and perform the **power stroke** — pulling thin filaments toward the center of the sarcomere
6. ATP binds myosin → head detaches → re-cocks for another stroke
7. Sarcomeres shorten across the whole fiber → muscle shortens → force is generated

Neuromuscular Junction and Motor Units

A **neuromuscular junction (NMJ)** is the synapse between a motor neuron and a muscle fiber. The neurotransmitter **acetylcholine (ACh)** diffuses across the synaptic cleft and triggers the muscle action potential. After the signal, the enzyme **acetylcholinesterase** rapidly breaks down ACh so the muscle can relax.

A **motor unit** = one motor neuron + all the muscle fibers it controls. Recruiting more motor units produces greater force. Small motor units (few fibers) allow fine control (eye muscles); large motor units (many fibers) generate powerful contractions (quadriceps).

Contraction Types

- **Isotonic contraction** — muscle changes length while tension is relatively constant; movement occurs *(e.g., picking up a glass, doing a bicep curl)*
- **Isometric contraction** — muscle generates tension but does **not** shorten; no movement *(e.g., holding a plank, carrying groceries without moving your arm)*

Muscle Fiber Types and Energy Systems

	Slow-Twitch (Type I)	**Fast-Twitch (Type IIa)**	**Fast-Twitch (Type IIx)**
Speed	Slow	Moderate-fast	Very fast
Fatigue resistance	Very high	Moderate	Low
Primary energy source	Aerobic (oxidative)	Mixed aerobic/ anaerobic	Anaerobic (glycolytic)
Best for	Endurance, posture	Sustained power	Short explosive bursts
Myoglobin / color	High (red)	Moderate	Low (white)
Example use	Marathon running	400 m swim	100 m sprint

Key energy concepts

- **ATP** is the immediate energy currency; muscles store only ~2–3 seconds worth
- **Creatine phosphate** replenishes ATP rapidly for ~10 seconds of intense activity

- **Anaerobic glycolysis** generates ATP quickly but produces **lactate** (and H^+), contributing to acidification and fatigue
- **Aerobic respiration** (using oxygen and mitochondria) is slower but far more efficient — it dominates during sustained activity in slow-twitch fibers

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Part 1: Concept Check — Before You Experiment

1. List the three types of muscle tissue and give one example location for each:

Muscle Type	Example Location

2. What is the role of calcium in muscle contraction? (Trace from nerve signal to cross-bridge formation in 3–4 steps.)

3. What is a motor unit, and how does increasing the number of recruited motor units affect the force of contraction?

4. What is the difference between an isotonic and an isometric contraction? Give one everyday example of each.

5. You hold a heavy bag of groceries without moving. What type of contraction are your arm muscles performing? What type of fiber (slow or fast twitch) would mainly be recruited for a 30-minute hold? Why?

Experiment 1: Grip Fatigue Test

Experiment 1 Background

When you repeatedly squeeze a muscle, it eventually fatigues — it produces less force or cannot sustain the same rate of contractions. Fatigue in fast-twitch fibers happens quickly (ATP and glycogen deplete, lactic acid builds). Slow-twitch fibers are more fatigue-resistant. This experiment measures how many squeezes you can complete in 30-second intervals, across 3 minutes total.

Experiment 1 Hypothesis

Before starting, predict what will happen to your squeeze count across the six 30-second intervals:

Experiment 1 Procedure

1. Sit comfortably with your dominant arm resting on the table, holding the stress/tennis ball.
2. Have a partner time six consecutive **30-second intervals** (3 minutes total). **Do not rest between intervals.**
3. In each interval, squeeze the ball as many times as you can at a comfortable but consistent pace.
4. Count and record your squeezes for **each 30-second interval**.
5. Repeat with your **non-dominant hand** if time allows.

Experiment 1 Data Table

Grip Fatigue — Dominant Hand

#	Interval	Time (min:sec)	Squeeze Count	Running Total
1				
2				
3				
4				
5				
6				

Grip Fatigue — Non-Dominant Hand (optional)

#	Interval	Time (min:sec)	Squeeze Count	Running Total
1				
2				
3				
4				
5				
6				

Analysis — Grip Fatigue

6. Calculate the percent change in squeeze count from Interval 1 to Interval 6 (dominant hand):

Percent change = (Interval 6 − Interval 1) ÷ Interval 1 × 100 = _____ %

7. Describe the trend in your data. Did your count decrease steadily, drop suddenly, or plateau? At which interval did you notice the biggest drop?

8. Explain the physiological mechanism behind the fatigue you observed. What is happening inside the muscle fibers (energy sources, waste products, calcium availability)?

9. The forearm muscles used in gripping contain a mix of slow-twitch and fast-twitch fibers. Based on your results, what does the rate of fatigue suggest about which fiber type tends to be recruited first — and which is exhausted first?

10. If you had rested for 2 minutes between intervals, predict how your results would differ. What physiological recovery processes would occur during that rest?

Experiment 2: Yardstick Drop — Measuring Reaction Time

Experiment 2 Background

Your nervous system controls muscle contraction, but there is always a measurable delay — **reaction time** — between detecting a stimulus (seeing something drop) and responding (catching it). This delay includes:

1. Sensory reception (light hits retina → signal sent)
2. Signal traveling to the brain
3. Brain processing and decision-making
4. Motor signal traveling to the muscles
5. Muscles contracting and the hand closing

We can estimate reaction time by measuring **how far a yardstick falls** before you catch it. Because objects fall under gravity at a predictable rate, we can convert the distance caught into time using:

> $t = \sqrt{(2d / g)}$
>
> *where* d *= drop distance in meters,* $g = 9.8\ m/s^2$

Quick reference table (for estimated conversions):

Distance caught (cm)	Reaction time (ms)
5	101
10	143
15	175
20	202
25	226
30	247

Experiment 2 Hypothesis

Before starting, predict: Will your reaction time improve over 10 trials, worsen, or stay roughly the same? Explain your reasoning in terms of nervous system function.

Experiment 2 Procedure

1. Sit at a table. Rest your **dominant hand** at the edge, palm facing in, fingers and thumb open (ready to pinch).
2. Your partner holds the yardstick vertically, with the **0-cm mark level with the top of your thumb**.
3. Your partner drops the yardstick **without warning**. Catch it as fast as you can by closing your fingers.
4. Record the **centimeter mark** at the top of your thumb at the catch point.
5. Repeat for **10 trials total**. Do not rush — wait for your partner to re-set between each trial.

Experiment 2 Data Table

Yardstick Drop — Reaction Time Trials

#	Trial	Distance dropped (cm)	Reaction time (ms) — calculate or look up
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Experiment 2 Summary Statistics

Reaction Time Summary

#	Statistic	Distance (cm)	Reaction Time (ms)
1			
2			
3			

Show your mean calculation here:

Experiment 2 Analysis — Reaction Time

11. Did your reaction time improve, worsen, or stay roughly the same across the 10 trials? Describe the trend you see in your data:

12. Identify your minimum (fastest) and maximum (slowest) trials. What is the range (max − min) in milliseconds? What factors might explain the trial-to-trial variability?

Range = _ ms

13. The yardstick drop measures the time for a sensory-to-motor loop through the nervous system. List the steps in order — from the drop to your hand closing — and identify WHERE in this chain most of the delay occurs:

14. Predict how each of the following would affect your reaction time, and explain the mechanism:

Factor	Prediction (faster / slower / no change)	Mechanism
Fatigue from the grip test before this experiment		
Practicing this test 50 times per day for a week		
Looking away from the yardstick instead of watching it		

Part 2: Synthesis

S1. In the grip fatigue test, your hand muscles were contracting isotonically. Describe what is happening at the level of the sarcomere during each squeeze. What happens to actin and myosin as you fatigue and ATP availability declines?

S2. Professional athletes in endurance sports (marathon runners, cyclists) have a higher proportion of slow-twitch fibers than sprinters. Based on what you observed in the grip fatigue test, describe the practical difference this fiber type distribution would make during a 3-hour race.

S3. The acetylcholine released at the neuromuscular junction must be broken down immediately after signaling so the muscle can relax. If a toxin (like certain nerve agents) blocked this breakdown, what would happen to the muscle? Connect your answer to the sliding filament mechanism.

S4. Looking back at the module questions and keys to success: which concept from the module (sliding filament theory, motor units, fiber types, neuromuscular junction, energy systems) did this lab make most concrete and real for you? Explain why.